

Two Scales in a Convention-Locked Preferential-Attachment Tree: Fixed-Vertex Martingales and the Global Degree Profile

HCS-C321 theorem-and-evidence package

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Abstract

For the linear preferential-attachment tree started from one edge, with no self-loops, we join two scales in one boundary-complete theorem. Every fixed vertex has exact rising-factorial moments of all orders and a moment-determinate square-root limit, whereas every fixed degree population converges in L^2 to $4/[k(k+1)(k+2)]$. The proof keeps fixed-vertex degree and degree counts separate. An exact finite computation independently audits the recurrences but is not used to infer either asymptotic theorem.

1 Frozen process and exact local law

Let T_2 consist of the single edge $\{1, 2\}$. Given T_n , add vertex $n+1$ and connect it to exactly one old vertex v , chosen with probability

$$\Pr(v \mid T_n) = \frac{d_v(n)}{2(n-1)}.$$

Thus there are no loops or multiple edges, n is the vertex-count clock, and $\sum_{v \leq n} d_v(n) = 2(n-1)$. Write $D_i(n) = d_i(n)$ for one fixed vertex, $N_k(n) = \#\{v \leq n : d_v(n) = k\}$ for a population, and

$$s_i = \begin{cases} 2, & i = 1, 2, \\ i, & i \geq 3. \end{cases}$$

This precise convention matters: several standard formulations use a self-loop or a linearized-chord-diagram seed [1, 2]. Fixed-vertex limit lineage is discussed by Backhausz [3]; the constants below are nevertheless derived for the declared one-edge process.

For $x^{\bar{r}} = x(x+1) \cdots (x+r-1)$, the identity $(d+1)^{\bar{r}} - d^{\bar{r}} = rd^{\bar{r}}/d$ gives

$$\mathbb{E}[D_i(n+1)^{\bar{r}} \mid T_n] = \left(1 + \frac{r}{2(n-1)}\right) D_i(n)^{\bar{r}}. \quad (1)$$

Theorem 1 (Exact all-order degree moments). *For every integer $r \geq 1$ and $n \geq s_i$,*

$$\mathbb{E}D_i(n)^{\bar{r}} = r! \frac{\Gamma(n-1+r/2)}{\Gamma(n-1)} \frac{\Gamma(s_i-1)}{\Gamma(s_i-1+r/2)}. \quad (2)$$

Proof. Every vertex has degree one at its birth time, so its initial rising factorial is $r!$. Iterating (1) and converting the finite product to a gamma quotient proves (2). $\square$

2 The fixed-vertex limit

Put

$$a_{s,n} = \prod_{t=s}^{n-1} \left(1 + \frac{1}{2(t-1)}\right).$$

Equation (1) at $r = 1$ makes $M_i(n) = D_i(n)/a_{s_i,n}$ a nonnegative martingale.

Theorem 2 (Square-root limit and determinacy). *For every fixed i , there is a nonnegative Z_i such that $D_i(n)/\sqrt{n} \rightarrow Z_i$ almost surely and in L^p for every finite p . For each integer $r \geq 1$,*

$$\mathbb{E}Z_i^r = \frac{r! \Gamma(s_i-1)}{\Gamma(s_i-1+r/2)}. \quad (3)$$

These moments determine the law of Z_i .

Proof. For every integer q , $D^q \leq D^{\bar{q}}$ and Theorem 1 gives $\mathbb{E}D_i(n)^q = O(n^{q/2})$. Since $a_{s_i,n} \asymp \sqrt{n}$, $(M_i(n))$ is bounded in every L^q . Martingale convergence and uniform integrability, choosing an integer $q > p$, give almost-sure and every-finite- L^p convergence. Moreover

$$\frac{a_{s,n}}{\sqrt{n}} \longrightarrow \frac{\Gamma(s-1)}{\Gamma(s-1/2)},$$

which transfers the convergence to $D_i(n)/\sqrt{n}$.

The difference $D^{\bar{r}} - D^r$ has degree at most $r - 1$ in D . Divide its expectation by $n^{r/2}$ and use the lower-order moment bounds; then apply gamma asymptotics to (2). Uniform integrability at order r follows from the order- $r + 1$ bound, proving (3). Finally, for $m_{2r} = \mathbb{E}Z_i^{2r}$, Stirling's formula gives

$$m_{2r}^{-1/(2r)} = \left(\frac{(2r)! \Gamma(s_i - 1)}{\Gamma(s_i - 1 + r)} \right)^{-1/(2r)} \geq c_i r^{-1/2}$$

for all sufficiently large r . Carleman's sum diverges, proving moment determinacy. $\square$

3 The global population scale

Set $N_0(n) = 0$. The newborn contributes to class one, and selection of an old degree- $k - 1$ or degree- k vertex moves one vertex into or out of class k . Hence

$$\mathbb{E}[\Delta N_k \mid T_n] = \mathbf{1}_{\{k=1\}} + \frac{(k-1)N_{k-1}(n) - kN_k(n)}{2(n-1)}, \quad |\Delta N_k| \leq 1. \quad (4)$$

Theorem 3 (Global degree profile in L^2). *For every fixed $k \geq 1$,*

$$\frac{N_k(n)}{n} \rightarrow p_k = \frac{4}{k(k+1)(k+2)} \quad \text{in } L^2.$$

Furthermore $\sum_{k \geq 1} p_k = 1$ and $\sum_{k \geq 1} k p_k = 2$.

Proof. The equilibrium equation from (4) is $p_k = \mathbf{1}_{\{k=1\}} + [(k-1)p_{k-1} - k p_k]/2$. It gives $p_1 = 2/3$ and $p_k = (k-1)p_{k-1}/(k+2)$, hence the displayed formula.

For the L^2 assertion, define $\xi_{k,n+1} = \Delta N_k - \mathbb{E}[\Delta N_k \mid T_n]$ and $U_{k,n} = N_k(n) - n p_k$. Then $\mathbb{E}(\xi_{k,n+1} \mid T_n) = 0$, $|\xi_{k,n+1}| \leq 2$, and

$$U_{k,n+1} = a_n U_{k,n} + b_n U_{k-1,n} + r_n + \xi_{k,n+1}, \quad (5)$$

where $a_n = 1 - k/[2(n-1)]$, $b_n = (k-1)/[2(n-1)]$, and $r_n = (p_k - \mathbf{1}_{\{k=1\}})/(n-1)$.

We prove $\mathbb{E}U_{k,n}^2 = O_k(n)$ by induction on k , beginning at a finite $n_0(k)$; earlier times enter the constant. After conditioning and squaring (5), all cross terms with ξ vanish and its variance is at most four. Fix $0 < \eta < k/2$. Young's inequality gives

$$2|a_n U_{k,n}(b_n U_{k-1,n} + r_n)| \leq \frac{\eta}{n} U_{k,n}^2 + \frac{n}{\eta} (b_n U_{k-1,n} + r_n)^2.$$

For $k = 1$ the b_n term is absent. At the induction step, $\mathbb{E}U_{k-1,n}^2 \leq C_{k-1}n$ implies

$$\mathbb{E}(b_n U_{k-1,n} + r_n)^2 \leq 2b_n^2 C_{k-1}n + 2r_n^2 = O_k(n^{-1}).$$

Because $a_n^2 = 1 - k/n + O_k(n^{-2})$, enlarge $n_0(k)$ so that $a_n^2 + \eta/n \leq 1 - c_k/n$ for some $c_k > 0$. Thus $\mathbb{E}U_{k,n+1}^2 \leq (1 - c_k/n)\mathbb{E}U_{k,n}^2 + C_k$. An induction on n yields $\mathbb{E}U_{k,n}^2 \leq C'_k n$, and division by n^2 proves the claim. Finally the two sums follow from the telescoping identities

$$\sum_{k=1}^K p_k = 1 - \frac{2}{(K+1)(K+2)}, \quad \sum_{k=1}^K k p_k = 2 - \frac{4}{K+2}.$$

$\square$

Theorems 2 and 3 expose two distinct scales: each fixed old vertex is a square-root hub, while the empirical degree population has a normalized cubic-tail profile. This paper makes no claim about the maximum degree, joint hub laws, $m > 1$, or uniformity as k grows with n .

Revision certificate: both microscopic and macroscopic limits

This round adds the square-root martingale limit, Carleman determinacy, and the global L^2 degree profile with its explicit second-moment induction.

References

- [1] A.-L. Barabási and R. Albert, "Emergence of scaling in random networks," *Science* 286 (1999), DOI: 10.1126/science.286.5439.509.
- [2] B. Bollobás, O. Riordan, J. Spencer, and G. Tusnády, "The degree sequence of a scale-free random graph process," *Random Structures Algorithms* 18 (2001), DOI: 10.1002/rsa.1009.
- [3] Á. Backhausz, "Limit distribution of degrees in random family trees," *Electron. Commun. Probab.* 16 (2011), DOI: 10.1214/ECP.v16-1598.